

# SOC INVESTIGATION REPORTS

## Phishing Email Investigation & Brute-Force Attack Investigation

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*SOC L1 Investigation Documentation*

### Investigation Report 02 — Brute-Force Attack

**Investigation Type:** Brute-Force / Authentication Attack Investigation

**Primary Focus:** Authentication logs, source IP analysis, login patterns, affected server

**Alert Source:** SIEM / Authentication Monitoring Alert

**Typical Alert:** Multiple Failed Login Attempts / Possible Brute-Force Attack

**Severity:** Medium to High depending on successful login, account privilege, asset criticality, and attack volume

#### 1. Incident Summary

A potential brute-force attack was investigated after repeated failed authentication attempts were detected against a server in the gaming segment. The investigation focused on authentication logs, source IP activity, login patterns, affected accounts, and the targeted server to determine whether the activity represented an unauthorized access attempt.

#### 2. Tools Used

- SIEM — Used to receive the alert, search authentication events, correlate failed and successful logins, and identify related activity.
- Wazuh — Used for centralized log collection, security alerts, authentication monitoring, and event investigation.
- Linux Authentication Logs — Used to review failed and successful authentication attempts, usernames, timestamps, and source addresses.
- IP Reputation / Threat Intelligence — Used to check whether the source IP had a history or reputation associated with malicious activity.
- Firewall / Network Logs — Used when available to verify connection attempts and traffic from the source IP toward the affected server.

#### 3. Alert Received

The investigation started from a SIEM alert for multiple failed authentication attempts. The alert showed repeated login failures associated with the same source IP and targeted server/account. The alert was reviewed to confirm the event pattern and establish the initial investigation scope.

#### 4. Investigation Steps

1. Review the SIEM alert. Confirm the alert timestamp, affected server, username or account, source IP, authentication method, and number of failed attempts.
2. Review authentication logs. Examine events for repeated failed logins and identify the frequency, timing, and pattern of the attempts.
3. Identify the source IP. Determine whether the same IP generated multiple authentication failures and whether it targeted other accounts or systems.
4. Check for successful authentication. Search immediately before and after the failed attempts to determine whether the source IP eventually achieved a successful login.

5. Review the affected account. Check whether the targeted account was privileged, active, disabled, or associated with a legitimate user or service.
6. Check source IP reputation. Use approved threat-intelligence sources to determine whether the IP has known malicious or suspicious history.
7. Correlate network events. Review firewall, network, or endpoint logs where available to confirm connections from the source IP to the affected server.
8. Determine attack scope. Check whether other users, servers, or services were targeted from the same source IP or during the same timeframe.
9. Assess severity. Consider number of attempts, attack duration, successful authentication, account privilege, server criticality, and evidence of compromise.
10. Document and escalate. Record the source IP, timestamps, targeted account/server, event pattern, evidence, and recommended response. Escalate according to SOC procedures.

## **5. How the Brute-Force Attack Was Identified**

- The initial SIEM alert showed repeated failed authentication attempts against the targeted server.
- Authentication logs were reviewed to confirm the repeated login pattern and source IP.
- The source IP was correlated across available security logs to identify additional activity.
- Successful login events were specifically checked to determine whether the attack resulted in unauthorized access.
- The affected account and server were reviewed to determine the potential business and security impact.

## **6. Key Findings**

- Multiple failed authentication attempts were observed against the targeted server.
- The repeated failures and source IP activity were consistent with potential brute-force behavior.
- Authentication events were correlated to validate the alert and determine the scope of activity.
- A successful authentication check was performed to determine whether the incident progressed beyond attempted access.

## **7. Response / Action Taken**

- Documented the source IP, affected server/account, timestamps, and authentication-event pattern.
- Escalated the incident when the activity met the required severity or showed evidence of successful unauthorized access.
- Recommended blocking or restricting the confirmed malicious source IP where appropriate and authorized.
- Recommended monitoring the affected account and server for additional authentication or post-compromise activity.

## **8. Investigation Outcome**

The investigation identified suspicious repeated authentication failures and established the source, target, event pattern, and potential scope of the activity. The collected evidence provided the required context for SOC triage, documentation, and escalation.